

IF440: Artificial Intelligence

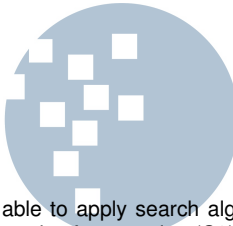
Topic 10: Certainty Factor

Moeljono Widjaja, Ph.D.
Yaya Suryana, Ph.D.

Universitas Multimedia Nusantara
Jl. Scientia Boulevard, Gading Serpong
Tangerang, Banten-15811 Indonesia

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Course Sub Learning Outcomes (Sub-CLO):

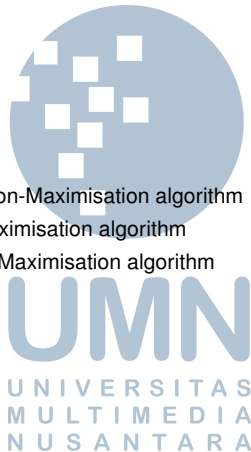


SubCLO-0221: Students are able to apply search algorithms, knowledge-based reasoning, probabilistic reasoning and soft computing (C3).



Review

- 1 Introduction to Expectation-Maximisation algorithm
- 2 Steps of Expectation-Maximisation algorithm
- 3 Example of Expectation-Maximisation algorithm



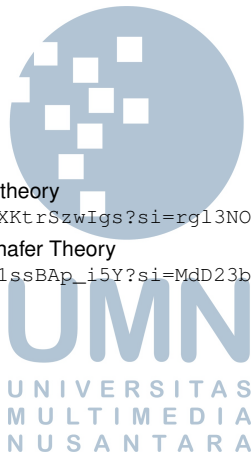
Pre-Class Assignments

- Lec-24 Dempster shafer theory

<https://youtu.be/UXKtrSzwIgs?si=rql3NO8fMDP9oEpz>

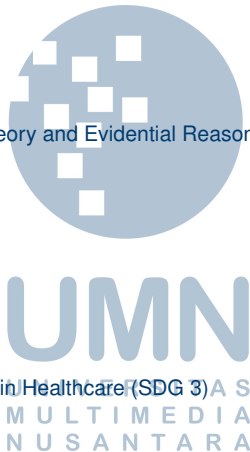
- Theory 11 - Dempster-Shafer Theory

https://youtu.be/5lssBAp_i5Y?si=MdD23bhwtPZz8oTS



Outline

- 1 Introduction
- 2 Certainty Factors (CF) Theory and Evidential Reasoning
- 3 Examples of CFs Theory
- 4 Dempster-Shafer Theory
- 5 Examples of DST
- 6 Application of DS Method in Healthcare (SDG 3)
- 7 Exercises





Introduction

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Introduction

Different types of imperfect information

■ Uncertainty

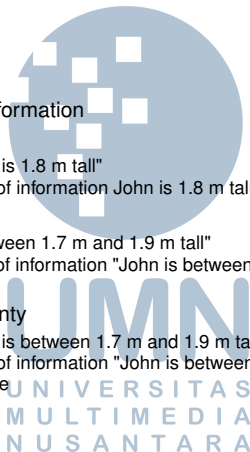
- Example: "I think John is 1.8 m tall"
- In this case, the piece of information John is 1.8 m tall is precise but uncertain

■ Imprecision

- Example: "John is between 1.7 m and 1.9 m tall"
- In this case, the piece of information "John is between 1.7 m and 1.9 m tall" is certain but imprecise

■ Imprecision and uncertainty

- Example: "I think John is between 1.7 m and 1.9 m tall"
- In this case, the piece of information "John is between 1.7 m and 1.9 m tall" is both uncertain and imprecise



BIAS OF BAYESIAN METHOD

- The framework of Bayesian reasoning requires **probability value as primary inputs**
- The assessment of these values usually involves **human judgement**
- However, psychological research shows that **humans cannot elicit probability values** consistent with the Bayesian rules
- This suggests that the conditional probabilities may be inconsistent with the **prior probabilities given by the expert**

BIAS OF THE BAYESIAN METHOD

- Consider, for example, a car does not start and makes odd noises when you press the starter. The conditional probability of the starter being faulty if the car makes odd noises may be expressed as :

IF the symptom is "odd noises"
 THEN the starter is bad {with probability 0.7}

$$\begin{aligned}
 p(\text{starter is not bad} \mid \text{odd noises}) &= \\
 &= p(\text{starter is good} \mid \text{odd noises}) = 1 - 0.7 = 0.3
 \end{aligned}$$

BIAS OF THE BAYESIAN METHOD

- Therefore, we can obtain a companion rule that states

IF	the symptom	is "odd noises"
THEN	the starter	is good {with probability 0.3}

- Domain experts do not deal with conditional probabilities and often deny the very existence of the hidden implicit probability (0.3 in our example)
- We would also use available statistical information and empirical studies to derive the following rules :

IF	the starter	is bad
THEN	the symptom	is "odd noises" {probability 0.85}

IF	the starter	is bad
THEN	the symptom	is not "odd noises" {probability 0.15}

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BIAS OF THE BAYESIAN METHOD

- To use the Bayesian rule, we still need the prior probability, the probability that the starter is bad if the car does not start. Suppose the expert supplies us the value of 5%. Now we can apply the Bayesian rule to obtain :

$$p(\text{starter is bad} | \text{odd noises}) = \frac{0.85 \cdot 0.05}{0.85 \cdot 0.05 + 0.15 \cdot 0.95} = 0.23$$

- The number obtained is significantly lower than the expert's estimate of 0.7!! (Given at the beginning of this section)
- The reason for the inconsistency is that the expert made different assumptions when assessing the conditional and prior probabilities

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Certainty Factors (CF) Theory and Evidential Reasoning

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Certainty Factors (CF) Theory and Evidential Reasoning

- Introduced by Shortliffe Buchanan on making MYCIN in 1975 to accommodate the expert's inexact reasoning.
- **MYCIN** was an early expert system that used artificial intelligence to identify bacteria causing severe infections.
- Doctors always analyze with "maybe", "probably", "almost certainly" so developers used CF to express the expert's degree of belief.
- Certainty Factor theory is a popular alternative to **Bayesian reasoning**.
- Uncertainty is represented as a **Degree of Belief**.
- Express the **Measure of Belief**.
- Manipulate degrees of belief when using knowledge-based systems.
- Certainty Factor express belief in an event based on evidence (or the expert's assessment)
 - +1.0 = absolute truth (complete confidence)
 - -1.0 = certain falsehood
- Certainty Factor are NOT probabilities.
- Certainty Factor need not sum to 1.

Uncertain Terms

Term	Certainty Factor
Definitely not	-1.0
Almost certainly not	-0.8
Probably not	-0.6
Maybe not	-0.4
Unknown	-0.2 to +0.2
Maybe	+0.4
Probably	+0.6
Almost certainly	+0.8
Definitely	+1.0

Certainty Factors (CF) Theory and Evidential Reasoning

- In inferencing with certainty factors, the knowledge base contains of a set of rules that have the following syntax:

```
IF <evidence>/<antecedent>
THEN <hypothesis>/<consequent> {cf}
```

- Where cf represents belief in hypothesis H given that evidence E has occurred.
- The certainty factors theory is based on two functions : measure of belief $MB(H,E)$ and measure of disbelief $MD(H,E)$.
- Measure of belief:** number that reflects the measure of increased belief in a hypothesis H based on evidence E . $0 \leq MB(H, E) \leq 1$
- Measure of disbelief:** number that reflects the measure of increased disbelief in a hypothesis H based on evidence E . $0 \leq MD(H, E) \leq 1$

Certainty Factors (CF) Theory and Evidential Reasoning

$$MB(H, E) = \begin{cases} 1 & \text{if } p(H) = 1 \\ \frac{\max(p(H|E), p(H)) - p(H)}{1 - p(H)} & \text{otherwise} \end{cases}$$

$$MD(H, E) = \begin{cases} 1 & \text{if } p(H) = 0 \\ \frac{\min(p(H|E), p(H)) - p(H)}{-p(H)} & \text{otherwise} \end{cases}$$

- $P(H)$ is the prior probability of hypothesis H being true
- $P(H/E)$ is the probability that hypothesis H is true given evidence E

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Certainty Factors (CF) Theory and Evidential Reasoning

- The values of $MB(H, E)$ and $MD(H, E)$ range between 0 and 1 . The strength of belief or disbelief in hypothesis H depend on the kind of evidence E observed. Some facts may increase the strength of belief, but some increase the strength of disbelief.
- The total strength of belief or disbelief in a hypothesis:

$$cf = \frac{MB(H, E) - MD(H, E)}{1 - \min[MB(H, E), MD(H, E)]}$$

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M U L T I M E D I A
N U S A N T A R A



Examples of CFs Theory

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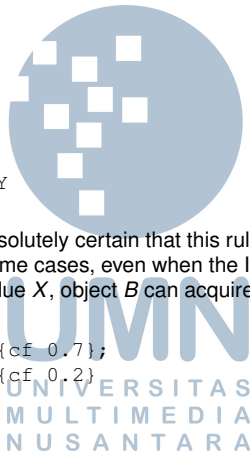
Examples of CFs Theory

- Consider a simple rule:

```
IF A is X
THEN B is Y
```

- An expert may not be absolutely certain that this rule holds. Also suppose it has been observed that in some cases, even when the IF part of the rule is satisfied and object *A* takes on value *X*, object *B* can acquire some different value *Z*.

```
IF A is X
THEN B is Y{cf 0.7};
      B is Z{cf 0.2}
```



CERTAINTY FACTORS (CF) PROPAGATION

- The certainty factor assigned by a rule is propagated through the reasoning chain. This involves establishing the net certainty of the rule consequent when the evidence in the rule antecedent is uncertain :

$$cf(H,E) = cf(E) \times cf$$

For example:

IF sky is clear
THEN the forecast is sunny {cf 0.8}

and the current certainty factor of sky is clear is 0.5, then $cf(H,E) = 0.5 \times 0.8 = 0.4$

This result can be interpreted as "It may be sunny".

CF FOR MULTIPLE RULE PREMISE

- For conjunctive rules such as :
 IF <evidence E_1 >
 :
 AND <evidence E_n >
 THEN <hypothesis H > {cf}

the certainty of hypothesis H , is established as follows:

$$cf(H, E_1 \cap E_2 \cap \dots \cap E_n) = \min [cf(E_1), cf(E_2), \dots, cf(E_n)] \times cf$$

For example:

IF	sky	is clear
AND	the forecast	is sunny
THEN	the action	is 'wear sunglasses' {cf 0.8}

and the certainty of **sky is clear** is **0.9** and the certainty of the forecast of **sunny** is **0.7**, then

$$cf(H, E_1 \cap E_2) = \min [0.9, 0.7] \times 0.8 = 0.7 \times 0.8 = 0.56 \text{ (probably)}$$

CF FOR MULTIPLE RULE PREMISE

- For disjunctive rules such as:

IF	<evidence E_1 >
	⋮
OR	<evidence E_n >
THEN	<hypothesis H > { cf }

the certainty of hypothesis H , is established as follows:

$$cf(H, E_1 \cup E_2 \cup \dots \cup E_n) = \max [cf(E_1), cf(E_2), \dots, cf(E_n)] \times cf$$

For example:

IF	sky	is overcast
OR	the forecast	is rain
THEN	the action	is 'take an umbrella' { cf 0.9}

and the certainty of **sky is overcast is 0.6** and the certainty of the **forecast of rain is 0.8**, then

$$cf(H, E_1 \cup E_2) = \max [0.6, 0.8] \times 0.9 = 0.8 \times 0.9 = 0.72 \text{ (almost certainly)}$$

CF SIMILARLY CONCLUDED RULES

- When the **same consequent** is obtained as a result of the execution of **two or more rules**, the individual certainty factors of these rules must be merged to give a combined certainty factor for a hypothesis.
- Suppose the knowledge base consists of the following rules:

Rule 1: IF A is X
 THEN C is Z {cf 0.8}

Rule 2: IF B is Y
 THEN C is Z {cf 0.6}

- What certainty should be assigned to object C having value Z if both Rule 1 and Rule 2 are fired?

CF SIMILARLY CONCLUDED RULES

- To calculate a combined certainty factor we can use the following equation :

$$cf(cf_1, cf_2) = \begin{cases} cf_1 + cf_2 \times (1 - cf_1) & \text{if } cf_1 > 0 \text{ and } cf_2 > 0 \\ \frac{cf_1 + cf_2}{1 - \min[|cf_1|, |cf_2|]} & \text{if } cf_1 < 0 \text{ or } cf_2 < 0 \\ cf_1 + cf_2 \times (1 + cf_1) & \text{if } cf_1 < 0 \text{ and } cf_2 < 0 \end{cases}$$

where:

cf_1 is the confidence in hypothesis H established by Rule 1;

cf_2 is the confidence in hypothesis H established by Rule 2;

$|cf_1|$ and $|cf_2|$ are absolute magnitudes of cf_1 and cf_2 , respectively.

CF SIMILARLY CONCLUDED RULES

- $cf(E1)=cf(E2)=1.0$
- $cf1(H,E1)=cf(E1) \times cf = 1.0 \times 0.8 = 0.8$
- $cf2(H,E2)=cf(E2) \times cf = 1.0 \times 0.6 = 0.6$
- $cf(cf1,cf2) = cf1(H,E1) + cf2(H,E2) * [1 - cf1(H,E1)]$
 $= 0.8 + 0.6 * (1 - 0.8) = 0.92$ (definitely)

Rule 1: IF A is X
 THEN C is Z {cf 0.8}

Rule 2: IF B is Y
 THEN C is Z {cf 0.6}



Dempster-Shafer Theory

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DEMPSTER – SHAFER THEORY

- Design to Deal with the distinction between uncertainty and ignorance.
- Rather than computing the probability of a proposition it computes **the probability that evidence supports the proposition**.
- DST is an evidence theory, it combines all possible outcomes of the problem. Hence it is used to solve problems where there may be a chance that a different evidence will lead to some different result.
- Consider a set of propositions as a whole
- Assign a set of propositions an interval [**believe, plausibility**] to constraint the degree of belief for each individual propositions in the set

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The Uncertainty in this Model

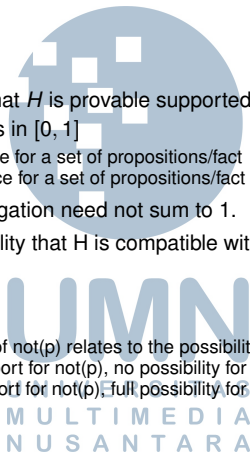
At its core, DST represents uncertainty using a mathematical object called a belief function. This belief function assigns degrees of belief to various hypotheses or propositions, allowing for a nuanced representation of uncertainty. Three crucial points illustrate the nature of uncertainty within this theory:

- 1 **Conflict:** In DST, uncertainty arises from conflicting evidence or incomplete information. The theory captures these conflicts and provides mechanisms to manage and quantify them, enabling AI systems to reason effectively.
- 2 **Combination Rule:** DST employs a combination rule known as Dempster's rule of combination to merge evidence from different sources. This rule handles conflicts between sources and determines the overall belief in different hypotheses based on the available evidence.
- 3 **Mass Function:** The mass function, denoted as $m(K)$, quantifies the belief assigned to a set of hypotheses, denoted as K . It provides a measure of uncertainty by allocating probabilities to various hypotheses, reflecting the degree of support each hypothesis has from the available evidence.

Source: <https://www.scaler.com/topics/artificial-intelligence-tutorial/dempster-shafer-theory-in-artificial-intelligence/>

BELIEF VS PLAUSIBILITY

- Belief is the probability that H is provable supported by the evidence.
- The belief measure bel is in $[0, 1]$
 - 0 – no support evidence for a set of propositions/fact
 - 1 – full support evidence for a set of propositions/fact
- Belief in a fact and its negation need not sum to 1.
- Plausibility is the probability that H is compatible with the available evidence (cannot be disapproved).
- The plausibility of p ,
 - $Pl(p) = 1 - Bel(not(p))$
 - Reflect how evidence of $not(p)$ relates to the possibility for belief in p .
 - $Bel(not(p))=1$: full support for $not(p)$, no possibility for p .
 - $Bel(not(p))=0$: no support for $not(p)$, full possibility for p .
 - Range is also in $[0,1]$.



Formal Definition

Frame of Discernment:

$$\Theta = \{A, B\}$$

Basic Belief Assignment (BBA):

$$m : 2^{\Theta} \rightarrow [0, 1], \quad m(\emptyset) = 0, \quad \sum_{A \subseteq \Theta} m(A) = 1$$

A mass function satisfies the following conditions:

- 1 $m(\emptyset) = 0$, meaning no belief is assigned to an empty set.
- 2 The sum of all assigned mass values must be 1, i.e., $\sum_{A \subseteq \Theta} m(A) = 1$.

Example mass functions:

$$m_1(A) = 0.6, m_1(B) = 0.2, m_1(A \cup B) = 0.2$$

$$m_2(A) = 0.5, m_2(B) = 0.3, m_2(A \cup B) = 0.2$$

Belief and Plausibility

Belief (Bel): total belief committed to A :

$$Bel(A) = \sum_{B \subseteq A} m(B)$$

Plausibility (Pl): degree to which A is not refuted:

$$Pl(A) = \sum_{B \cap A \neq \emptyset} m(B)$$

For $\Theta = \{A, B\}$:

$$Bel(A) = m(A), \quad Pl(A) = m(A) + m(A \cup B)$$

$$Bel(B) = m(B), \quad Pl(B) = m(B) + m(A \cup B)$$

Basic Belief Assignment Table



Evidence	$m(A)$	$m(B)$	$m(A \cup B)$
m_1	0.6	0.2	0.2
m_2	0.5	0.3	0.2



Dempster's Rule of Combination

$$m_{12}(C) = \frac{\sum_{A \cap B = C} m_1(A)m_2(B)}{1 - K}$$

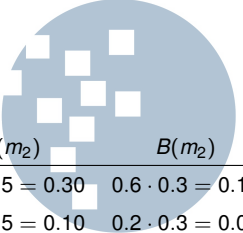
$$K = \sum_{A \cap B = \emptyset} m_1(A)m_2(B)$$

where K is a measure of the amount of conflict between the two mass sets.

Purpose: combine independent evidence and reduce uncertainty.

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Combination Matrix



	$A(m_2)$	$B(m_2)$	$A \cup B(m_2)$
$A(m_1)$	$0.6 \cdot 0.5 = 0.30$	$0.6 \cdot 0.3 = 0.18$	$0.6 \cdot 0.2 = 0.12$
$B(m_1)$	$0.2 \cdot 0.5 = 0.10$	$0.2 \cdot 0.3 = 0.06$	$0.2 \cdot 0.2 = 0.04$
$A \cup B(m_1)$	$0.2 \cdot 0.5 = 0.10$	$0.2 \cdot 0.3 = 0.06$	$0.2 \cdot 0.2 = 0.04$

Conflict:

$K = 0.18 + 0.10 = 0.28$

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Final Combined Belief Masses

$$m_{12}(A) = \frac{0.30 + 0.12 + 0.10}{0.72} = 0.7222$$

$$m_{12}(B) = \frac{0.06 + 0.04 + 0.06}{0.72} = 0.2222$$

$$m_{12}(A \cup B) = \frac{0.04}{0.72} = 0.0556$$

⇒ Evidence strongly favors A

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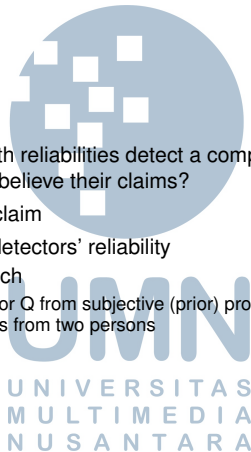


Examples of DST

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Example 1

- Two persons M and B with reliabilities detect a computer and claim the result independently. How you believe their claims?
- Question (Q): detection claim
- Related question (RQ): detectors' reliability
- Dempster-Shafer approach
 - Obtain belief degrees for Q from subjective (prior) probabilities for RQ for each person
 - Combine belief degrees from two persons



Example 1 (continue...)

■ Person M:

- reliability 0.9, unreliability 0.1
- Claim h_1
- Belief degree of h_1 is $bel(h_1) = 0.9$
- Belief degree of $not(h_1)$ is $bel(not(h_1)) = 0.0$, different from probability theory, since no evidence supporting $not(h_1)$
- $pl(h_1) = 1 - bel(not(h_1)) = 1 - 0 = 1$
- Thus belief measure for M claim h_1 is $[0.9, 1]$

■ Person B:

- Reliability 0.8, unreliability 0.2
- Claim h_2
- $bel(h_2) = 0.8, \quad bel(not(h_2)) = 0$
- $pl(h_2) = 1 - bel(not(h_2)) = 1 - 0 = 1$
- Belief measure for B claim h_2 is $[0.8, 1]$

Example 1 (continue...)

Set of propositions: M claim h_1 and B claim h_2

■ Case 1: $h_1 = h_2$

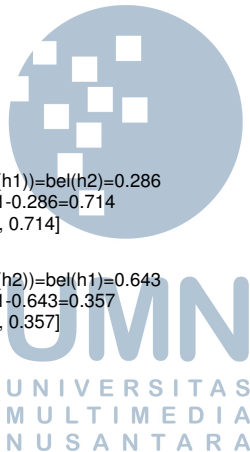
- Reliability M and B: $0.9 \times 0.8 = 0.72$
- Unreliability M and B: $0.1 \times 0.2 = 0.02$
- The probability that at least one of two is reliable: $1 - 0.02 = 0.98$
- Belief measure for $h_1 = h_2$ is $[0.98, 1]$

■ Case 2: $h_1 \neq h_2$

- Cannot be both correct and reliable
- At least one is unreliable
- Reliable M and unreliable B: $0.9 \times (1 - 0.8) = 0.18$
- Reliable B and unreliable M: $0.8 \times (1 - 0.9) = 0.08$
- Unreliable M and B: $(1 - 0.9) \times (1 - 0.8) = 0.02$
- At least one is unreliable: $0.18 + 0.08 + 0.02 = 0.28$
- Given at least one is unreliable, posterior probabilities
- Reliable M and unreliable B: $0.18 / 0.28 = 0.643$
- Reliable B and unreliable M: $0.08 / 0.28 = 0.286$

Example 1 (continue...)

- Belief measure for h1
 - $\text{Bel}(h1)=0.643$, $\text{bel}(\text{not}(h1))=\text{bel}(h2)=0.286$
 - $\text{Pl}(h1)=1-\text{bel}(\text{not}(h1))=1-0.286=0.714$
 - Belief measure: $[0.643, 0.714]$
- Belief measure for h2
 - $\text{Bel}(h2)=0.286$, $\text{bel}(\text{not}(h2))=\text{bel}(h1)=0.643$
 - $\text{Pl}(h2)=1-\text{bel}(\text{not}(h2))=1-0.643=0.357$
 - Belief measure: $[0.286, 0.357]$



Example 2

Suppose two experts are consulted regarding a system failure. The failure could be caused by Component A, Component B or Component C. The first expert believes that the failure is due to Component A with a probability of 0.99 or Component B with a probability of 0.01 (denoted by $m_1(A)$ and $m_1(B)$, respectively). The second expert believes that the failure is due to Component C with a probability of 0.99 or Component B with a probability of 0.01 (denoted by $m_2(C)$ and $m_2(B)$, respectively). The distributions can be represented by the following:

■ Expert 1:

- $m_1(A) = 0.99$ (failure due to Component A)
- $m_1(B) = 0.01$ (failure due to Component B)

■ Expert 2:

- $m_2(B) = 0.01$ (failure due to Component B)
- $m_2(C) = 0.99$ (failure due to Component C)

Example 2 (continue ...)

The combination of the masses associated with the experts is summarized in this table.

			Expert 1			
			A	B	C	Failure Cause
			0.99	0.01	0	m_1
Expert 2	Failure Cause	m_2				
	A	0	$m_1(A)m_2(A) = 0$	$m_1(B)m_2(A) = 0$	$m_1(C)m_2(A) = 0$	
	B	0.01	$m_1(A)m_2(B) = 0.0099$	$m_1(B)m_2(B) = 0.0001$	$m_1(C)m_2(B) = 0$	
	C	0.99	$m_1(A)m_2(C) = 0.9801$	$m_1(B)m_2(C) = 0.0099$	$m_1(C)m_2(C) = 0$	

Example 2 (continue ...)

- 1 To calculate the combined basic probability assignment for a particular cell, simply multiply the masses from the associated column and row.
- 2 Where the intersection is nonempty, the masses for a particular set from each source are multiplied, e.g., $m_{12}(B) = (0.01)(0.01) = 0.0001$.
- 3 Where the intersection is empty, this represents conflicting evidence and should be calculated as well. For the empty intersection of the two sets A and C associate with Expert 1 and 2, respectively, there is a mass associated with it. $m_1(A)m_2(C) = (0.99)(0.99) = (0.9801)$.
- 4 Then sum the masses for all sets and the conflict.
- 5 The only nonzero value is for the combination of B , $m_{12}(B) = 0.0001$. In this example there is only one intersection that yields B , but in a more complicated example it is possible to find more intersections to yield B .
- 6 For K , there are three cells that contribute to conflict represented by empty intersections.

$$K = (0.99)(0.01) + (0.99)(0.01) + (0.99)(0.99) = 0.9999$$
- 7 Calculate the joint, $m_1(B)m_2(B) = (.01)(.01)/[1 - 0.9999] = 1$



Application of DS Method in Healthcare (SDG 3)

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A Web-Based Expert System for Diagnosing Cervical Cancer Using Dempster-Shafer Method

A Web-Based Expert System for Diagnosing Cervical Cancer Using Dempster-Shafer Method

Publisher: IEEE

[Cite This](#)



Alvin Martin Djong ; Moeljono Widjaja [All Authors](#)

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Abstract

Document Sections

I. Introduction

II. Theories

III. Research Methods

IV. Result and Discussion

V. Conclusion

[Authors](#)

[Figures](#)

[References](#)

Abstract:

Cervical cancer is classified among the diverse array of cancer types that have been identified, primarily attributed to the infection of Human Papillomavirus (HPV). Cervical cancer is a form of cancer that can be effectively prevented. However, patients frequently initiate treatment after the cancer has metastasized and progressed to an advanced stage, rendering it challenging to manage and contributing to its elevated mortality rate. Therefore, a cervical cancer diagnostic expert system has been developed utilizing the Dempster-Shafer method on the web. This system aims to facilitate the early detection of cervical cancer by analyzing symptoms and risk factors. The findings of this study demonstrate the successful development of an expert system application that utilizes the Dempster-Shafer method to generate results. The application achieved a high level of accuracy, specifically 93.34 %, as determined by the diagnosis conducted by an expert specializing in cervical cancer. The application also collects user satisfaction data, which reveals an average score of 89.50% based on the End-User Computing Satisfaction (EUCS) method. This indicates a high level of satisfaction among the majority of users when utilizing the application.

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Advantages and Disadvantages of Dempster Shafer Theory

■ Advantages

- 1 **Better uncertainty handling:** Unlike classical probability, DST allows partial belief assignment, making it useful in situations where complete probability distributions are unavailable.
- 2 **Ideal for AI applications:** DST is widely used in AI-driven decision-making, sensor fusion, and expert systems, where multiple uncertain sources of evidence need to be combined.
- 3 **No need for prior probabilities:** Unlike Bayesian inference, DST does not require strict prior probability distributions, making it flexible for real-world AI applications.

■ Disadvantages

- 1 **Computational complexity:** As the number of possible hypotheses increases, DST's calculations become computationally expensive, making it difficult to scale in large AI models.
- 2 **Defining mass functions:** The effectiveness of DST depends on properly defining mass functions, which can be challenging without sufficient domain expertise or reliable evidence sources.

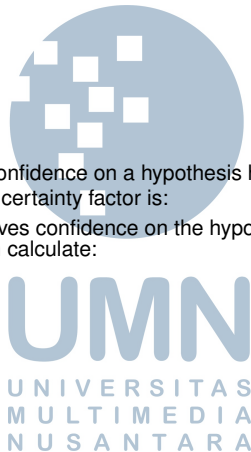


Exercises

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Exercise 1: CF

- 1 If an observation gives confidence on a hypothesis $h1$ with $MB[h1,e] = 0.5$ and $MD[h1,e] = 0.2$, then the certainty factor is:
- 2 If the observation also gives confidence on the hypothesis $h2$ with $MB[h2,e] = 0.8$ and $MD[h2,e] = 0.1$, then calculate:
 - $cf[h1 \wedge h2, e]$
 - $cf[h1 \vee h2, e]$



Exercise 2: CF

- Given the following:

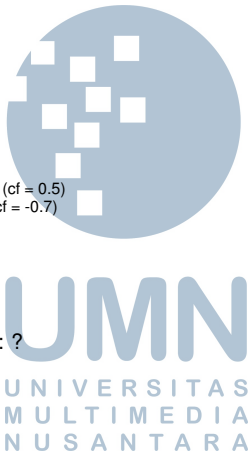
- Rule Base:

- R1: if a and b then x (cf = 0.5)
 - R2: if c or d then x (cf = -0.7)

- Dynamic Input:

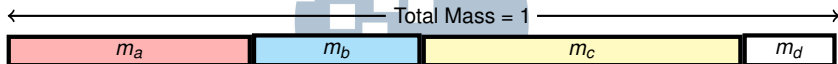
- a, with certainty 1.0
 - b, with certainty 0.8
 - c, with certainty 0.9
 - d, with certainty 0.7

- Compute CF values for x: ?



Exercise 3: DST

Witness: "I'm fairly sure that the car was either brown or black. Probably black, though it could have been brown. I could be wrong though."



- $m_a = 0.3$: Car was black.
- $m_b = 0.2$: Either black or brown.
- $m_c = 0.4$: Car was brown.
- $m_d = 0.1$: Neither.
- Apply the DST method to calculate the belief measure for the following hypotheses:
 - h1: The car was black.
 - h2: The car was brown.

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Exercise 4: DST

- Frame of discernment $\Theta = \{A, B\}$.

Two independent evidence sources give mass functions m_1 and m_2 :

$$m_1(\{A\}) = 0.60, \quad m_1(\{B\}) = 0.20, \quad m_1(\Theta) = 0.20.$$

$$m_2(\{A\}) = 0.50, \quad m_2(\{B\}) = 0.30, \quad m_2(\Theta) = 0.20.$$

(Each mass to subsets sums to 1.)

- The matrix form

Evidence	$m(A)$	$m(B)$	$m(A \cup B)$
m_1	0.6	0.2	0.2
m_2	0.5	0.3	0.2

Questions:

- 1 Based on only evidence m_1 , calculate the belief measure for the hypotheses A and B .
- 2 Based on only evidence m_2 , calculate the belief measure for the hypotheses A and B .
- 3 Based on both evidences m_1 and m_2 , calculate the belief measure for the hypotheses A and B .

References

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- [1] A. M. Djong and M. Widjaja, "A Web-Based Expert System for Diagnosing Cervical Cancer Using Dempster-Shafer Method," in **2023 7th International Conference on New Media Studies (CONMEDIA)**. IEEE, 12 2023, pp. 243–249. [Online]. Available: <https://ieeexplore.ieee.org/document/10428193/>
- [2] G. Shafer, "Probability Judgment in Artificial Intelligence and Expert Systems," <https://doi.org/10.1214/ss/1177013426>, vol. 2, no. 1, pp. 3–16, 2 1987. [Online]. Available: <https://projecteuclid.org/journals/statistical-science/volume-2/issue-1/Probability-Judgment-in-Artificial-Intelligence-and-Expert-Systems/10.1214/ss/1177013426.full>

U N I V E R S I T A S
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Next Week

- Introduction to Fuzzy Logic
 - Crisp Sets
 - Fuzzy Set Theory
 - Operations on Fuzzy Sets
 - Membership Functions
- Fuzzy Inference System
 - Fuzzification
 - Defuzzification
 - Fuzzy Rules
- Exercise



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- 1 Organizing learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practicing informatics science and technology society.

